THE MINIMAL VERSAL FORMAL OBJECT OF A PREDEFORMATION CATEGORY SATISFYING (S2) HAS BIJECTIVE ORBITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the diagram interpretation through the vector space structure, the k-linearity of the differential and the action by addition, the construction of the subring on which the derivation vanishes with its cotangent space computation, and the final contradiction with minimality via the lifting lemma.

Editorial note. The label lemma-construct-bijective-orbits names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-construct-bijective-orbits –start 4743 –end 4834 –local-id 28012 returned RESULT=OK (L28012 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json (second exclusive pool assigned by the Lead on 2026-10-13).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Existence and minimality of a versal formal object. Same chapter, the lemmas producing a minimal versal formal object of the predeformation category. Supplies the minimal versal object xi over R used throughout the proof and the minimality property invoked at the end.
- Vector space structure and linearity of the differential. Same chapter, the lemma giving the tangent space its k-vector space structure, the lemma that the differential of a formal object is k-linear, and the lemma identifying the action of the derivations of k with addition. Supply the interpretation of the displayed diagram: the kernel K, the linearity of the differential and the action by addition.
- Lifting of a section along a morphism hitting the trivial deformation. Same chapter, the lemma allowing a morphism x -> y to be produced once the pushforward of the trivial object is reached. Supplies the morphism x -> y obtained from the triviality of x_epsilon.
- Descending the versal object along a nonsurjective ring map. Same chapter, part (1) of the lemma characterizing the smallest ring over which a versal

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object descends. Supplies the contradiction with minimality once the ring map R'/m_R^2 -> R/m_R^2 is seen to be nonsurjective.

Source reference. Original equation 06T6: sources/stacks-formal-defos/pinned-original-source.tex, source lines 4623–4623. Prerequisite pointer invoked at proof line(s) 4747; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 06S6: sources/stacks-formal-defos/pinned-original-source.tex, source lines 808–808. Prerequisite pointer invoked at proof line(s) 4775; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06SU: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3811–3811. Prerequisite pointer invoked at proof line(s) 4763; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06H0: sources/stacks-formal-defos/pinned-original-source.tex, source lines 887–887. Prerequisite pointer invoked at proof line(s) 4782; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06IJ: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3716–3716. Prerequisite pointer invoked at proof line(s) 4760; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06IS: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3212–3212. Prerequisite pointer invoked at proof line(s) 4824; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06T5: sources/stacks-formal-defos/pinned-original-source.tex, source lines 4558–4558. Prerequisite pointer invoked at proof line(s) 4752; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06T2: sources/stacks-formal-defos/pinned-original-source.tex, source lines 4375–4375. Prerequisite pointer invoked at proof line(s) 4831; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06IH: sources/stacks-formal-defos/pinned-original-source.tex, source lines 3669–3669. Prerequisite pointer invoked at proof line(s) 4757; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{F}$ be a predeformation category satisfying (S2) which has a versal formal object. Then its minimal versal formal object satisfies (06T6).*

Proof. Let ξ be a minimal versal formal object for $\mathcal{F}$, see Lemma 06T5. Say ξ lies over $R \in \mathrm{Ob}(\widehat{\mathcal{C}}_\Lambda)$. In order to parse (06T6) we point out that $T\mathcal{F}$ has a natural k-vector space structure (see Lemma 06IH), that $d\underline{\xi} : \mathrm{Der}_\Lambda(R,k) \to T\mathcal{F}$ is linear (see Lemma 06IJ), and that the action of $\mathrm{Der}_\Lambda(k,k)$ is given by addition (see Lemma 06SU). Consider the diagram

$$\begin{array}{ccccc} & & \mathrm{Hom}_k(\mathfrak{m}_R/\mathfrak{m}_R^2, k) & & \\ & & \uparrow & & \\ K & \longrightarrow & \mathrm{Der}_\Lambda(R,k) & \xrightarrow{d\underline{\xi}} & T\mathcal{F} \\ & & \uparrow & \nearrow & \\ & & \mathrm{Der}_\Lambda(k,k) & & \end{array}$$

The vector space K is the kernel of $d\underline{\xi}$. Note that the middle column is exact in the middle as it is dual to the sequence (06S6). If (06T6) fails, then we can find a nonzero element $D \in K$ which does not map to zero in $\mathrm{Hom}_k(\mathfrak{m}_R/\mathfrak{m}_R^2, k)$. This means there exists an $t \in \mathfrak{m}_R$ such that $D(t) = 1$. Set $R' = \{a \in R \mid D(a) = 0\}$. As D is a derivation this is a subring of R. Since $D(t) = 1$ we see that $R' \to k$ is surjective (compare with the proof of Lemma 06H0). Note that $\mathfrak{m}_{R'} = \mathrm{Ker}(D : \mathfrak{m}_R \to k)$ is an ideal of R and $\mathfrak{m}_R^2 \subset \mathfrak{m}_{R'}$. Hence

$$\mathfrak{m}_R/\mathfrak{m}_R^2 = \mathfrak{m}_{R'}/\mathfrak{m}_R^2 + k\bar{t}$$

which implies that the map

$$R'/\mathfrak{m}_R^2 \times_k k[\epsilon] \to R/\mathfrak{m}_R^2$$

sending ϵ to $\bar{t}$ is an isomorphism. In particular there is a map $R/\mathfrak{m}_R^2 \to R'/\mathfrak{m}_R^2$.

Let $\xi \to y$ be a morphism lying over $R \to R/\mathfrak{m}_R^2$. Let $y \to x$ be a morphism lying over $R/\mathfrak{m}_R^2 \to R'/\mathfrak{m}_R^2$. Let $y \to x_\epsilon$ be a morphism lying over $R/\mathfrak{m}_R^2 \to k[\epsilon]$. Let x_0 be the unique (up to unique isomorphism) object of $\mathcal{F}$ over k. By the axioms of a category cofibred in groupoids we obtain a commutative diagram

$$\begin{array}{ccc} y & \longrightarrow & x_\epsilon \\ \downarrow & & \downarrow \\ x & \longrightarrow & x_0 \end{array} \quad \text{lying over} \quad \begin{array}{ccc} R'/\mathfrak{m}_R^2 \times_k k[\epsilon] & \longrightarrow & k[\epsilon] \\ \downarrow & & \downarrow \\ R'/\mathfrak{m}_R^2 & \longrightarrow & k. \end{array}$$

Because $D \in K$ we see that x_ϵ is isomorphic to $0 \in \mathcal{F}(k[\epsilon])$, i.e., x_ϵ is the pushforward of x_0 via $k \to k[\epsilon], a \mapsto a$. Hence by Lemma 06IS we see that there exists a morphism $x \to y$. Since $\mathrm{length}_\Lambda(R'/\mathfrak{m}_R^2) < \mathrm{length}_\Lambda(R/\mathfrak{m}_R^2)$ the corresponding ring map $R'/\mathfrak{m}_R^2 \to R/\mathfrak{m}_R^2$ is not surjective. This contradicts the minimality of ξ/R, see part (1) of Lemma 06T2. This contradiction shows that such a D cannot exist, hence we win. □